

MD. ABU HORAIRA AL REZON

rezon.2205011@bau.edu.bd | +880 1799 144 788 | Mymensingh, BD

[Portfolio](#) | [ResearchGate](#) | [LinkedIn](#) | [GitHub](#)

Research-driven Agricultural Engineering student and CIGR 2025-26 Gold Winner specializing in Precision Agriculture, with experience in Data Science, Computer Vision and IoT-based Agricultural Automation. Passionate about developing affordable, scalable, and data-driven solutions for sustainable farming.

EDUCATION

- **B.Sc in Agricultural Engineering** March 2022 - Present
Department of Farm Power and Machinery
Bangladesh Agricultural University, Mymensingh
Expected Graduation: June 2027
- **Higher Secondary Certificate** March 2018 - March 2020
Sherpur Govt. College, Sherpur, BD
GPA: 5.00 / 5.00

CORE TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++
- **Technologies:** OpenCV, Open3D, Scikit-learn, TensorFlow, Pandas, NumPy, IoT, Linux, Git/GitHub
- **Software Tools:** ArcGIS, QGIS, Google Earth Engine, PIX4D, FreeCAD, Fusion 360, MATLAB
- **Hardware:** Arduino, Raspberry Pi, Drone/UAV, Vis-NIR Spectroscopy
- **Languages:** Bangla (Native), English (Fluent)

PROJECTS & TECHNICAL WORK

- **Prediction of Early Blight Disease in Potato using Machine Learning** Link | 2025
 - Developed a practical and scalable ML-based solution to detect and manage early blight in potato crops.
 - Enabled precise, data-driven decision-making for farmers and promoted sustainable agricultural practices.
- **AgriSpectro: Affordable Spectroscopic Device for Non-destructive Fruit Quality Analysis** Link | 2025
 - Engineered a low-cost Vis-NIR spectroscopic device to non-destructively assess internal fruit quality.
 - Addressed postharvest losses of 25-50% in major Bangladeshi fruits by enabling on-site quality assessment.

RESEARCH INTERESTS

Agricultural Automation and Robotics | AI-driven Precision Crop Monitoring and Management | Food Quality Assessment and Postharvest Technology | Climate-smart Farming Systems

PROFESSIONAL EXPERIENCE

- **Ambassador (Business Competition)** March 2026 - Present
Smart Agro-Technology Innovation Youth Network - SAIYN
 - Promoting youth innovation through national and international agri-business competitions.
- **Co-founder & CTO** September 2025 - Present
Bluvia | A Startup Offering 100% Organic and Natural Herbal Beverages
 - Served as official Refreshment Partner at the 3rd National Career Carnival organized by BAUCC at BAU.
 - Operated a beverage stall at the Tea Utshob 3.0, organized by Team Utshob at BAU.
- **Executive Member** August 2025 - Present
Bangladesh Agricultural University Career Club - BAUCC
 - Contributed to organizing the 3rd National Career Carnival at Bangladesh Agricultural University (BAU).

ACHIEVEMENTS & ACTIVITIES

- First (Gold Winner) at International Student Competition 2025-26 | CIGR Certificate | 2026
- Pre-seed Winner at Cohort 3 | University Innovation Hub Program Certificate | 2025
- Second at Farm Mechanization Idea Competition 2025 | BARI Certificate | 2025

CERTIFICATIONS & LICENSES

- Geo-spatial Analysis using ArcGIS, Google Earth Engine, and Python | Geo Lense Certificate | 2025
- Agricultural Entrepreneurship Development Workshop | iFarmer & FlyMek Certificate | 2025